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WEEK 03

Cybersecurity Fundamentals & Security Concepts

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Task 1 - Protocols and Port Numbers

1. Write 15 common network protocols with their default port numbers.

Protocol	Port Number(s)	Transport	Purpose
HTTP	80	TCP	Unencrypted web browsing
HTTPS	443	TCP	Secure/encrypted web traffic (SSL/TLS)
SSH / SFTP	22	TCP	Secure remote login and file transfer
FTP	20, 21	TCP	File Transfer Protocol (data and control)
DNS	53	TCP/UDP	Resolves domain names to IP addresses
SMTP	25	TCP	Sends email between servers
IMAP	143	TCP	Retrieves email while leaving it on the server
POP3	110	TCP	Retrieves email and downloads it to the client
DHCP	67, 68	UDP	Dynamically assigns IP addresses to devices
RDP	3389	TCP	Remote Desktop Protocol for Windows
Telnet	23	TCP	Unencrypted remote text communications/login
NTP	123	UDP	Network Time Protocol for clock synchronization
LDAP	389	TCP/UDP	Lightweight Directory Access Protocol for directory services
SNMP	161	UDP	Simple Network Management Protocol for device monitoring
TFTP	69	UDP	Trivial File Transfer Protocol for simple/fast transfers

Task 2 - HTTP vs HTTPS Analysis

2. Visit 10 different websites and analyze the following

Answer:

HTTP: HTTP stands for hyper text transfer protocol - It is a protocol that a web browser needs to transfer/transmit the data. The data here is in the form of a plain text and that data can be read by anyone in between. Example: Sending a postcard. Anyone who gets it can read the message.

HTTPS: HTTPS stands for hyper text transfer protocol secure - It is a secure version of HTTP and standard for modern web applications which is used to transfer/transmit data securely, before the data is transferred the data is SSL/TLS encrypted with this encryption the data cannot be read by anyone in between. Example: Sending a sealed package. The contents are protected while they're in transit.

Website	Website uses HTTP or HTTPS?	Automatically redirect from HTTP to HTTPS	Is the connection secure
google.com	HTTPS	Yes	Yes (TLS 1.3, HSTS Enforced)
wikipedia.org	HTTPS	Yes	Yes (TLS 1.3, HSTS Enforced)

amazon.com	HTTPS	Yes	Yes (TLS 1.2/1.3)
github.com	HTTPS	Yes	Yes (TLS 1.3, HSTS Enforced)
apple.com	HTTPS	Yes	Yes (TLS 1.3)
microsoft.com	HTTPS	Yes	Yes (TLS 1.3)
youtube.com	HTTPS	Yes	Yes (TLS 1.3)
linkedin.com	HTTPS	Yes	Yes (TLS 1.3)
bbc.com	HTTPS	Yes	Yes (TLS 1.2/1.3)
cnn.com	HTTPS	Yes	Yes (TLS 1.2/1.3)
nasa.gov	HTTPS	Yes	Yes (TLS 1.3, Federal Mandate)

Commands	Website1	
ping	google.com	
nmap	scanme.nmap.org	Your own local IP address
dig	google.com	github.com
nslookup	google.com	github.com

[illegible]

Nmap scanme.nmap.org

```
(ayeshakhan@AmnaKhan)-[~]
$ nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-17 00:41 +0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp   open  nping-echo
31337/tcp  open  Elite

Nmap done: 1 IP address (1 host up) scanned in 5.31 seconds

(ayeshakhan@AmnaKhan)-[~]
$
```

Nmap Your own local IP address

```
(ayeshakhan@AmnaKhan)~$ nmap
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-19 01:39 +0500
Nmap scan report for 39.47.41.133
Host is up (0.0056s latency).
Not shown:
PORT      STATE      SERVICE
Nmap done: scanned in 1.92 seconds
```

Dig

```
ayeshakhan@AmnaKhan: ~  
ayeshakhan@AmnaKhan: ~  
Name: github.com  
Address: 20.207.73.82  
  
(ayeshakhan@AmnaKhan)-[~]  
$ dig google.com  
  
;<>> DiG 9.20.23-1-Debian <>> google.com  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<- opcode: QUERY, status: NOERROR, id: 56117  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
;; QUESTION SECTION:  
;google.com. IN A  
  
;; ANSWER SECTION:  
google.com. 102 IN A 142.250.202.142  
  
;; Query time: 7 msec  
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)  
;; WHEN: Fri Jul 17 00:59:29 PKT 2026  
;; MSG SIZE rcvd: 55  
  
(ayeshakhan@AmnaKhan)-[~]  
$ dig github.com  
  
;<>> DiG 9.20.23-1-Debian <>> github.com  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<- opcode: QUERY, status: NOERROR, id: 44744  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
;; QUESTION SECTION:  
;github.com. IN A  
  
;; ANSWER SECTION:  
github.com. 6 IN A 20.207.73.82  
  
;; Query time: 15 msec  
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)  
;; WHEN: Fri Jul 17 00:59:43 PKT 2026  
;; MSG SIZE rcvd: 55
```


Nslookup

```
(ayeshakhan@AmnaKhan)~  
$ nslookup google.com  
Server:          10.255.255.254  
Address:         10.255.255.254#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.250.202.142  
Name:   google.com  
Address: 2a00:1450:4019:80f::200e  
  
(ayeshakhan@AmnaKhan)~  
$ nslookup github.com  
Server:          10.255.255.254  
Address:         10.255.255.254#53  
  
Non-authoritative answer:  
Name:   github.com  
Address: 20.207.73.82
```


Task 4 - Research Task – What Happens When You Type a URL and Press Enter?

When a URL (Uniform Resource Locator) is typed into a web browser and as soon the Enter button is hit, a complex, high-speed sequence of networking protocols and rendering engines are executed in the background. The following main operations are performed at the backend. All these steps happen each time we enter any URL and all these processes happen in the background and within milliseconds.

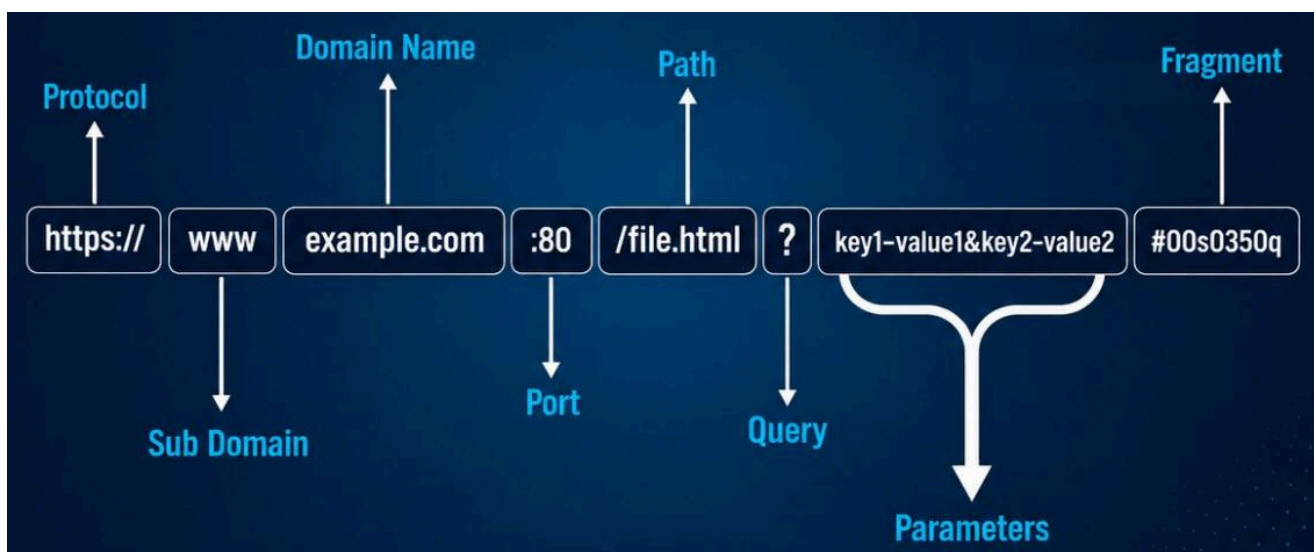
Enter the URL:

1. DNS resolution
2. TCP three-way handshake
3. HTTPS upgrade
4. HTTP Request/Response
5. Server Response
6. Browser rendering the webpage

Here is the complete step-by-step process of what happens behind the scenes.

Enter the URL

When you type a URL (Uniform Resource Locator) like `https://example.com:80/file.html?key1=value1&key2=value2#00s0350q`, the browser starts interpreting it:



- **Protocol:** https:// tells the browser how to communicate (secure HTTP here).
- **Domain Name:** example.com is the domain name of the site. It is the memorable address and points to a specific server's IP address.
- **Path & Query:** /products?id=123 tells the server which resource you want.

1. DNS Lookup and IP Address Resolution

Computers communicate using IP addresses (like 192.0.2.1), but humans use readable domain names (like example.com). The Domain Name System (DNS) bridges this gap.

Domain Name System (**DNS**) is a data store that stores the mapping from a domain name to its IP address. Just like an address book or a telephone book.

The DNS checks at the following places for the IP address:

Local Caching: The browser first checks its own DNS cache. If the IP isn't there, it checks the Operating System's cache, then the router's cache, and finally the local ISP's recursive DNS server.

The DNS Hierarchy: If the IP is still not found, the ISP's server queries the DNS hierarchy: the Root Name Server, the Top-Level Domain (TLD) server (e.g., .com), and finally the Authoritative Name Server for the specific domain, which returns the correct IP address.

Result: The domain name is translated into an IP address.

2. TCP Three-Way Handshake

Once the browser has the target IP address, it must establish a reliable connection with the web server. Since web traffic uses the Transmission Control Protocol (TCP), it initiates a three-way handshake. It ensures both browser and server are ready to communicate:

1. **SYN (Synchronize):** (Browser says hello) - The client sends a packet with the SYN flag set, asking the server if it is open for new connections.
2. **SYN-ACK (Synchronize-Acknowledge):** (Server replies hello)
The server responds with a SYN-ACK packet, confirming it received the request and is ready.
3. **ACK (Acknowledge):** (Browser confirms) - The client sends an ACK packet back, establishing a stable connection over port 443 (for HTTPS) or port 80 (for HTTP).

3. Secure Connection (TLS Handshake for HTTPS)

- **ClientHello / ServerHello:** Browser and server agree on encryption protocols.
- **Certificate Exchange:** They exchange certificates to verify trust (via Certificate Authorities).
- **Key Exchange:** A secure encrypted channel is created.

Result: This ensures data you send/receive is private.

4. HTTP Request

Now that the browser is connected to the server, the browser sends an HTTP request to the server for the page's contents. The server can determine what the client wants to do using the information in the request line.

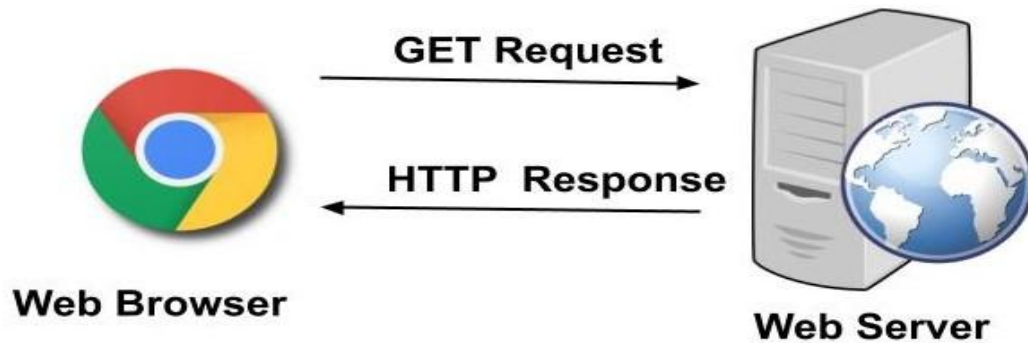
```
GET / HTTP/1.1
Host: example.com
User-Agent: Chrome/120.0
Accept: text/html
```

This tells the server: “Hey, I want the homepage of example.com.”

5. Response from the server

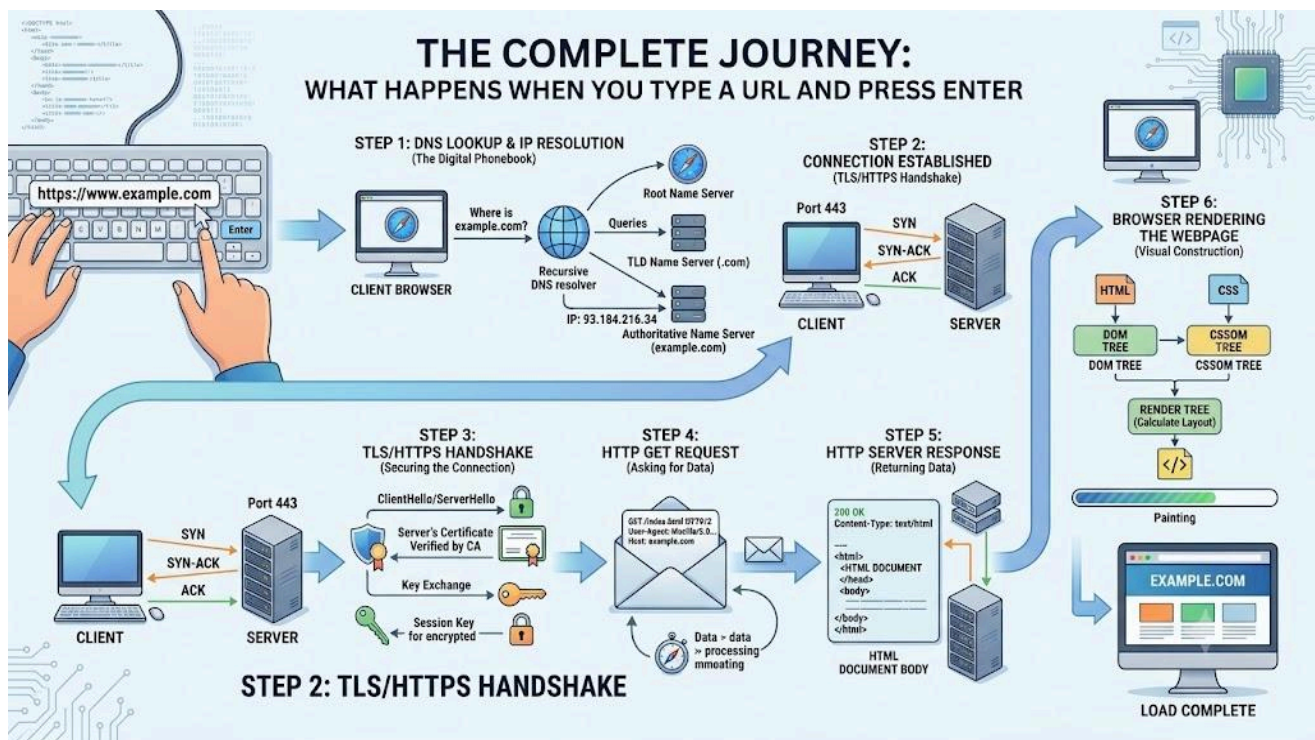
On the server side:

- The request reaches the **web server software** (like Nginx, Apache, Node.js).
- The server may fetch data from **databases**, run **application logic**, or return a **static file**.
- The server responds with an **HTTP response**, e.g.: followed by the HTML of the page.



6. Browser rendering the response from the server

Now the browser gets the response and the HTML web page is rendered in phases. First, it gets the HTML structure and then it sends multiple GET requests to get the embedded links, images, CSS, javascript files, etc and renders the final webpage we see.



Task 5 - Networking Cheat Sheet

 NETWORKING PROTOCOLS AND PORTS 				
 PROTOCOL	 PORT NUMBER(S)	 TRANSPORT PROTOCOL	 DESCRIPTION	
 HTTP	80	TCP	Web traffic	
 HTTPS	443	TCP/UDP (HTTP/3 QUIC)	Secure web traffic	
 FTP (Control)	21	TCP	File transfer control	
 FTP (Data Active)	20	TCP	File transfer data (active mode)	
 FTP (Passive)	Dynamic Ports	TCP	File transfer (passive mode)	
 SFTP	22	TCP	Secure file transfer over SSH	
 SSH	22	TCP	Secure remote login	
 Telnet	23	TCP	Insecure remote login (deprecated)	
 SMTP	25	TCP	Email sending	
 SMTPS	465	TCP	Secure email sending	
 POP3	110	TCP	Email retrieval	
 POP3S	995	TCP	Secure email retrieval	
 IMAP	143	TCP	Email retrieval (advanced)	
 IMAPS	993	TCP	Secure email retrieval	
 DNS	53	TCP/UDP	Domain name resolution (UDP mainly)	
 DHCP Server	67	UDP	Assigns IP addresses	
 DHCP Client	68	UDP	Receives IP address	
 TFTP	69	UDP	Trivial file transfer	
 SNMP	161	UDP	Network management	
 SNMP Trap	162	UDP	SNMP alerts/traps	
 LDAP	389	TCP	Directory services	
 LDAPS	636	TCP	Secure LDAP	
 MSSQL	1433	TCP	Database service	
 Oracle DB	1521	TCP	Database service	

Task 6 - Kali Linux Web Enumeration

Use the following tools to analyze the given websites:

Tools	Website1	Website 2
whatweb	google.com	github.com
whois	google.com	github.com

Whatweb - google.com

```
(ayeshakhan@AmnaKhan)~$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.200.174], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only, X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.200.174], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID,Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.152.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only, X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]]
https://www.google.com/ [200 OK] Cookies[AEC,NID,Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.155.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
```

Whatweb - github.com

```
(ayeshakhan@AmnaKhan)~$ whatweb github.com
http://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
```

Whois - google.com

```

ayeshakhan@AmnaKhan: ~
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:00Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-15T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-16T19:49:28Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure

```

Whois - github.com

```

ayeshakhan@AmnaKhan: ~
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-4021.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-16T19:53:13Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not

```